

Unit 1

This session will be recorded



Unit 1 summative assessment

- [This](#) is the assessment handbook – ***all of the criteria for the project is in here and is the primary resource you should refer to.***
- The rest of this slide-deck will go through the assessment handbook and provide context throughout

Project overview

- In this project, you will build a project combining Python packages for Data Analysis and Data Visualisations for fast Data Analytics prototyping. This project allows you to apply Python tools to carry out basic data preparation tasks and demonstrate a foundational understanding of how analytics supports evidence-based decision-making in real-world contexts.
- The project immerses you in an environment that fully reflects professional business requirements. It achieves this by encouraging you to reflect on the “whys” and “hows” of a data analysis system that delivers tangible value to your organisation. The data analysis is conducted in accordance with the business requirements. You will need to use your skills in data extraction, transformation, and loading, as well as data visualisation using Python tools.

Project ideas

- Project Idea 1: Retail Sales Data Analysis
- Project Idea 2: Healthcare Insurance Cost Analysis
- Project Example Idea 3: Credit Card Customer Churn Analysis
- Project Example Idea 4: Online Retail Transaction Analysis
- Project Example Idea 5: Car Price Analysis

I will create groups in Discord so that you can have study groups. Everyone must submit their own work but there is nothing wrong with helping each other along the way.

Learning Outcomes

[Here](#) is a simple checklist which you can refer to

General assessment criteria

To achieve these criteria, keep the following aspects in mind when creating your project.

- The learner has a clear rationale for the development of this project and has produced a fully functioning, **well-documented Jupyter Notebook** for a real-life audience.
- The finished project has a **clear, well-defined purpose** addressing a particular target audience (or multiple related audiences). Its purpose would be immediately evident to a new user without having to look at supporting documentation.
- The **code is well-organised and easy to follow**. The development process is evident through commit messages. The project's documentation provides a clear rationale for its development and covers all stages of the development life cycle.

General assessment criteria

- The learner has documented a **clear, justified rationale** for a real-world application and a comprehensive explanation of how it will be developed.
- The finished project is judged to be publishable in its current form.
- It clearly matches the design and demonstrates the characteristics of craftsmanship in the code. The **resulting application is original** and not a copy of any walkthrough projects encountered in the course.
- The learner must elaborate, **either in a README or a Notebook** and show how their analysis supports evidence-based decision-making in their real-world context.
- The learner **must display multiple different plot types in the Notebook**, each helping to answer a business requirement.

Development and Implementation

- Code demonstrates characteristics of '**clean code.**'
- **Consistent and appropriate naming conventions** within code and in file naming, e.g. File names and class names are descriptive and consistent.

Robust code

- **No logic errors** are found when running the code.
- The full design is implemented, providing a good solution to the users' demands and expectations.

Version control systems are used effectively

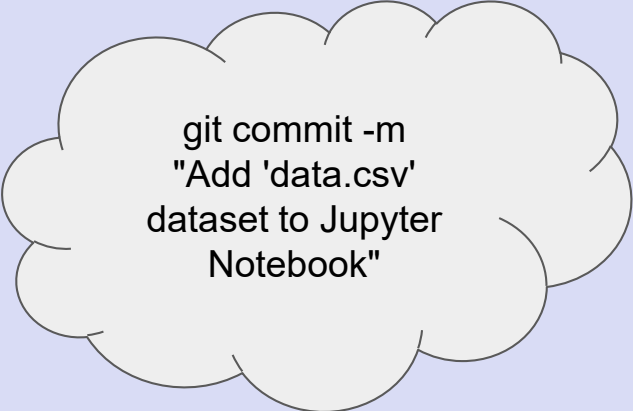
- **All code is managed in git with commit messages that reflect the specific reason for the commit.**
- There is a **separate, well-defined commit for each individual feature/fix.**
- There are **no very large commits**, which makes it harder to understand the development process and could lead the assessor to suspect plagiarism.

Commit messages

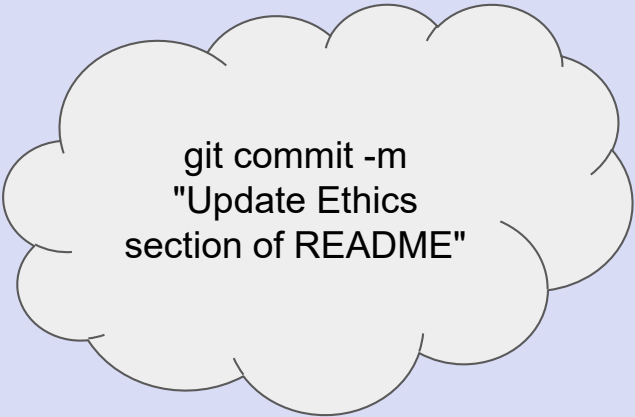
- Keep it short and concise – **50 characters or less**
- Use the 'imperative mood' (e.g., "Add feature" not "Added feature")
- Common starters:
 - "Add..."
 - "Debug..."
 - "Fix..."
 - "Update..."

Commit messages

- Little and often is the best^{[L][SEP]}
- Every time you add something new^{[L][SEP]}
- You can't 'overcommit' but you can 'undercommit'!^{[L][SEP]}
- 'git push' every so often



```
git commit -m  
"Add 'data.csv'  
dataset to Jupyter  
Notebook"
```



```
git commit -m  
"Update Ethics  
section of README"
```

The development process is documented

- **The purpose of the application is clearly described** – key project goals, target audience...

README

- Recommended structure for a README.md file: [Template](#)
- Your project can include a readme file named README.md, in markdown format, that will describe all aspects of your project.
- If you include a README, it must be in English. We strongly recommend that your Notebook should be in English, too. The assessment will be conducted in English, so as to ensure that the functionality and navigation can be fully assessed, we recommend that the site be in English.

Project 1 specific considerations

- You **don't need to include** 'Dashboard Design' and 'Deployment'. [L]
[SEP]
You will learn about these for future projects.
- You will go into more depth about 'Ethical Considerations' in Unit 3. At this stage you should include one or two ethical considerations to show that you have thought about it. You don't need to spend a long time on it though.

Markdown language

- README documentation and Jupyter notebooks notes should be written in 'Markdown'.
- If you see a file name ending name.MD then that file is written in Markdown.
- It's not dissimilar to writing in any other word processing system, it just has certain syntax elements that you need to use. See this cheatsheet: <https://www.markdownguide.org/cheat-sheet/>

Plagiarism

- Plagiarism, as defined by the Oxford Dictionary, is “the practice of taking someone else's work or ideas and passing them off as one's own.” It is a serious academic offence for which there are serious consequences.
- **It is acceptable to use and *reference* others' code**; however, it is an academic plagiarism offence if any piece of work that is not entirely the student's own is not correctly referenced or acknowledged. All student projects submitted will be reviewed for plagiarism. This includes checking code comparison tools, plagiarism software, and review of git commit history, and other mechanisms.

Plagiarism

- **It is the responsibility of each student to ensure that any direct or indirect inclusion of the work of others is fully and adequately acknowledged.** We appreciate that plagiarism may be unintentional. However, it will still be treated as an offence. The Facilitation team can answer any plagiarism-related queries, but as a general rule, if in doubt, include attribution of all sources.
- Students are encouraged to ask mentors, tutors and their peers for advice about their project work but any submission should not include any code written by others unless it is explicitly credited to them. Failure to correctly credit code that a student hasn't created themselves will be considered plagiarism and will result in a failing grade. Blatant or repeated offences of plagiarism will not be tolerated and will result in stringent penalties being applied, including removal from the course.

Submission

Unit 1 Summative Assessment

-  Learning Outcome understanding
-  Create Capstone repo from template
-  Assessment Guide
-  Plagiarism
-  Presentations and Report
-  Submit Capstone here

Submission form questions

Prepare answer for these questions in advance of submitting your project:

1. Briefly describe how you have used AI to build your project.
2. Briefly describe the challenges you faced when using AI to build your project.
3. Briefly describe how AI benefited you and your project.
4. Explain the ethical considerations taken into account when building your project.

Previous projects

- https://github.com/DaniDL346/Global_Methane_Flux_Analysis
- https://github.com/banerjexplores/plant_health-monitor
- <https://github.com/MahaSci/heart-disease-data-hackathon-1>
- <https://github.com/Adam-Ansar/London-Crime>
- https://github.com/Iqra-qbl/Analysis_of_diabetes_risk-
- <https://github.com/JayneLawley/Project-1-Healthcare-Insurance-Cost>
- <https://github.com/Old-Rusty/global-gdp-and-co2-emissions>
- <https://github.com/Ninestein666/Healthcare-Insurance-Cost-Analysis-Assessment>
- https://github.com/Sonya-J00/Chronic_Kidney_Disease
- https://github.com/Hamza-Murshad/Electronic_Sales

Creating repository and
project board



Use this template

- [Template](#)
- You must use [Python 3.12.8](#) with this template.

Helpful guides

- [DA – setting up a project board](#)
- [GitHub & VS Code](#)
- [Solo Git Branching Working Practice](#)
- [Git Salvation](#)